

[VOL. 03 // 2026]

Dad jokes inside MongoDB shell

who needs data analysis when you can have
memes?

SIH 2026

01

STUPID IDEAS HACKATHON 2016

WEEKEND EDITION

```
> db.jokes.find()
```

Dad Jokes

— Inside —

MongoDB Shell

```
> Why don't skeletons fight each other?
```

```
✂ Because they don't have the guts! :-)
```

```
■
```



Agenda

- WhoAmI
- Some context behind this project
- Demo time!
- [If some time left] Tech implementation details

Dad joke to start: If you decide to store all these jokes they should really go in a dad-a-base.



WhoAmI

- No, I'm not from uWaterloo :(Graduated as Software Engineer major back in 2020 back in Kyiv, Ukraine
- Developing Azure DocumentDB
- Developing databases since 2019 with some gap (was developing AI services for Xbox, got burned out, return back to good old databases)
- I'm obsessed with memes (maybe I need to check with my shrink on this)

I still hope I will get admitted to MMath in Stats next year though



Some context behind this project



The Ghost of Christmas Past

Back in 2020 I was adding support of ARRAY data type for database called SingleStore (fka MemSQL) and I thought to myself - we need a MEME data type support to store, query and ensemble queries with memes



the Ghost of Christmas Present

saw SIH stand on Socratica Symposium 2026 and understood it's time to execute. Currently in db space it's popular to have text to query AI usage where you describe your query in natural language language and you get your DB query from AI.



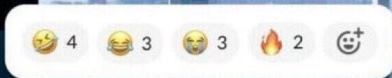
the Ghost of Christmas Yet to Come

But who needs text-to-query feature for real? me? nope
What you actually is some lame jokes about your data and higher total compensation 🤖

Demo time!



“paula” @paularambles · 8h



35

3K

27K

1.1M



05

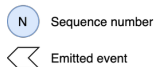
“

> LMFAO
my VP

Tech implementation details

- Pull request is on GH: [AndrewKhoma/mongosh/pull/1](#)
- Used Copilot plugin Phased Agent Workflow
- Used LLM model Claude Opus 4.6 (with 1M context) to code
- Used Azure OpenAI instance (with gpt-5-chat model) for jokes generation
- What I've learnt during this hackathon:
 - this presentation tool (lol), wanted to try it like forever
 - read about REPL (Read-Evaluate-Print loop) and how it's implemented in MongoDB shell
 - was successful in hacking it :)

CLI code evaluation



- Async Rewriter transforms synchronous-looking user input (e.g. `db.coll.find({})`) into code with auto-inserted `await` statements so shell API promises resolve transparently

- Shell Evaluator orchestrates the pipeline: detects legacy commands (show dbs) for direct execution, otherwise passes JS input through the async rewriter then evaluates it against the Shell API context

- `MongoshNodeRepl` → `ShellEvaluator` → `AsyncRewriter`
→ `eval()` is the execution chain — user input flows down, gets rewritten, gets evaluated by Node's REPL, and results bubble back up

